

# Nabhit Arorra

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## EDUCATION

College of Engineering | The Pennsylvania State University | State College, PA

B.S in Computational Data Science | Minor in Computer Science and Statistics

May-26

Relevant Course: Machine Learning and Algorithmic AI, Data Structures and Algorithms, Applied Statistics Parallel Computing, Object Oriented Programming, Discrete Mathematics, R Programming, Applied Multivariate Statistical Analysis

## TECHNICAL SKILLS

- **Programming & Scripting:** Python | Java | C++ | C# | R | SQL | Scala
- **Data & ML Frameworks:** TensorFlow | Keras | PyTorch | Deep Learning | CNNs | Natural Language Processing | Computer Vision
- **Big Data & Cloud:** Hadoop (HDFS, MapReduce) | Spark (RDDs, DataFrames) | AWS (EC2, S3) | Git/GitHub
- **Statistical & Data Analysis:** ggplot2 | tidyverse | broom | SQL queries | Data Wrangling
- **Operating Systems & Dev Tools:** Linux | Windows | macOS | RStudio | ServiceNow | MS Power Apps

## EXPERIENCE

Student Technician | Penn State IT Services | Erie, PA

[Link](#)

Jan-23 to May-24

- Engineered an advanced Inventory Management Application using MS Power Apps to streamline asset tracking and enhance operational efficiency, leveraging ServiceNow for real-time support ticketing.
- Spearheaded the installation, configuration, and maintenance of PCs, printers, and peripherals across Linux, Windows, and macOS environments, ensuring optimal performance.
- Directed technology storage and inventory processes, introducing data-driven controls to improve organization and distribution.

Co-Founder & Coding Lead | Quillian-AI | Bangalore, India

[Link](#)

Jun-20 to Jul-22

- Created a deep learning model achieving 97% accuracy for early melanoma detection using CNNs (TensorFlow/Keras/PyTorch) with advanced data augmentation on ISIC dermatology datasets.
- Implemented ResNet and Xception architectures, applying hyperparameter tuning to mitigate overfitting and improve real-world reliability.
- Collaborated with Tata Memorial Hospital to ensure clinical validation, aligning machine learning pipelines with healthcare standards.

Team Leader, The Junior Academy | New York Academy of Sciences | New York, NY

Oct-20 to Jun-22

- Led AI-driven sustainability and telemedicine solutions, including lab-grown cotton (70% soil reduction, 90% water reduction) and blockchain-based supply chain/transparency frameworks.
- Integrated TensorFlow and Keras into a telemedicine platform for symptom-based doctor allocation, enforcing data privacy with blockchain to secure patient records.

## PROJECTS

- **Social Media Analytics:** Constructed a distributed pipeline using Hadoop (HDFS, MapReduce) and Spark (RDDs, DataFrames) for near-real-time text analysis on large-scale social media data, employing Scala, Git, and CI/CD to streamline cluster deployment for sentiment analysis and topic modeling.
- **Morse Code Translator:** Developed a Python CLI tool to convert messages to and from Morse code, emphasizing robust error handling and test-driven development, and employing modular architecture (logging, unit tests) to demonstrate core software engineering principles.
- **Reddit Topic Modeling:** Built an NLP pipeline in Python using LDA to classify posts across 11 subreddits, leveraging TF-IDF, word embeddings, and text preprocessing (tokenization, stopword removal), then minimized multi-class logloss through hyperparameter tuning.
- **Predicting Antibodies with XGBoost:** Created a supervised ML pipeline to forecast Omicron antibody presence, performing extensive feature engineering (demographics, exposure variables) and parameter optimization, and incorporating SHAP-based interpretability to achieve low RMSE.
- **CVS Store Sales Forecasting:** Developed a time-series regression model in Python to predict weekly department-level sales by engineering lag features, integrating holiday and regional data, and implementing cross-validation to guide data-informed inventory and staffing decisions.

## CERTIFICATIONS

- **Amazon Web Services**, Data Analytics Certifications (in progress)
- **Harvard University**, Technology Entrepreneurship – Lab to Market
- **Stanford University**, Machine Learning
- **Duke University**, Programming Foundations
- **Xelerate Academics**, Artificial Intelligence
- **Microsoft Engineering**, Undergraduate & Masters Asia Virtual Experience program